

SAHIL BHOIR

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[Github](#) • [Linkedin](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

Results-driven Computer Engineering graduate (CGPA 9.477) with a minor in Artificial Intelligence and Machine Learning from Vidyalankar Institute of Technology, Mumbai. Built production-ready AI systems including a real-time violence detection platform (YOLOv8, 20+ FPS), an image captioning pipeline (BLEU-4 ~0.22), a serverless AWS receipt processor (92% OCR accuracy), and a deployed NLP web application. AWS Certified Cloud Practitioner and IBM Full Stack Developer. Seeking a graduate or entry-level role in AI/ML Engineering, Cloud Development, or Full-Stack Software Engineering.

PROJECTS

Real-Time Violence Detection System – [Demo Website](#)

2026

Computer Vision • YOLOv8 • FastAPI • Streamlit • Docker • Twilio

- Designed and deployed an end-to-end AI surveillance system detecting violent activities (Fighting, Weapon, Aggression) from live video streams using a fine-tuned YOLOv8 model at 15–30 FPS.
- Implemented multi-channel alert dispatch via WhatsApp (Twilio) and Email (SMTP) within 5 seconds of a confirmed detection event, including automatic screenshot capture.
- Built temporal consistency filtering to eliminate false positives and reduce spurious alerts in continuous video streams.
- Developed a 4-page Streamlit monitoring dashboard with real-time confidence scores, detection history logs, and Docker deployment for portability.

Serverless Receipt Processor - [Website](#)

2024

AWS Lambda • S3 • Textract • DynamoDB • API Gateway • Step Functions

- Built a fully serverless, event-driven AWS application that processes receipt and invoice images using Amazon Textract OCR, achieving 92%+ accuracy extracting vendor names, amounts, and dates.
- Designed scalable architecture with S3 event triggers, Lambda auto-scaling (500+ concurrent requests), Step Functions orchestration, and DynamoDB storage — delivering sub-5 second average processing time.
- Implemented intelligent rule-based auto-categorisation supporting 8+ expense categories with per-user data isolation in DynamoDB; reduced manual processing effort by 80%+.

NLP Multi-Feature Text Analyzer - [Website](#)

2024

Python • Flask • spaCy • Streamlit • HuggingFace • displaCy

- Developed a deployed NLP platform (HuggingFace Spaces) supporting sentiment analysis, summarization, named entity recognition, question answering, and keyword extraction for 5,000+ word documents.
- Optimised inference pipelines using spaCy and transformer-based models, reducing average processing latency by 37% while improving entity extraction accuracy; API response time under 800ms.

Image Captioning — CNN + LSTM Encoder-Decoder – [Demo Website](#)

2025

PyTorch • ResNet50 • LSTM • GloVe • Flickr8k

- Built an image captioning system using a ResNet50 CNN encoder and 2-layer LSTM decoder with GloVe 200-dim embeddings, trained on the Flickr8k dataset (8,000 images, 40,000 captions).
- Achieved BLEU-4 ~0.22 and METEOR ~0.23 on the Flickr8k test split using beam search decoding (k=5); implemented teacher forcing during training to stabilize convergence.
- Evaluated model quality at corpus level using NLTK BLEU and NLTK METEOR metrics; implemented both greedy and beam search inference pipelines for comparison.

LSTM-Based Predictive Maintenance System

2024

PyTorch • LSTM • Time-Series • Scikit-learn

- Designed and trained stacked and bidirectional LSTM models in PyTorch to predict Remaining Useful Life (RUL) from multivariate sensor data, achieving RMSE of 12.4 and MAE of 9.8.
- Engineered a sliding-window preprocessing pipeline that improved prediction accuracy ~28% over baseline models; built interactive visualisations demonstrating proactive fault detection.

TECHNICAL SKILLS

Languages: Python (Expert), JavaScript, Java, C/C++

AI / Machine Learning: PyTorch, TensorFlow/Keras, Scikit-learn, spaCy, NLTK, HuggingFace Transformers, LangChain, LSTM, Time-Series Forecasting, Pandas, NumPy

Computer Vision: YOLOv8/Ultralytics, OpenCV, ResNet50, CNN, Object Detection, Image Segmentation, Real-Time Inference

Cloud & Backend: AWS (Lambda, S3, EC2, DynamoDB, Textract, API Gateway, Step Functions, EventBridge, CloudWatch), Flask, FastAPI, Streamlit, REST API Design, Node.js

DevOps & Tools: Docker, Kubernetes, Git/GitHub, Linux, CI/CD Pipelines, Terraform (beginner)

Frontend: React.js, HTML/CSS, Tailwind CSS, JavaScript

EDUCATION

Bachelor of Engineering — Computer Engineering

2021 – 2025

Vidyalankar Institute of Technology, Mumbai • *Minor: Artificial Intelligence & Machine Learning*

CGPA: 9.477 / 10.0

HSC — Narayana Junior College, Kalyan

81%

SSC — Jindal Vidya Mandir, Vasind

96.2%

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services

IBM Full Stack Software Developer Professional Certificate — IBM / Coursera (12 courses: React, Node.js, Docker, Kubernetes, CI/CD)

AWS Cloud Practitioner Essentials — Amazon Web Services

Machine Learning in Python — Udemy

ADDITIONAL

Currently upskilling in: Agentic AI Architectures, LLM Fine-Tuning, Advanced MLOps, AWS Solutions Architect

Interests: Agentic AI, RAG Systems, Real-Time Computer Vision, MLOps, Cloud-Native Architectures